

PRACTICAL WEEK-5

TASK1: Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
#include <time.h>

int main() {
    int fd[2];
    pid_t pid;
    char buffer[100];

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid > 0) {
        // Parent - Producer
        close(fd[0]);

        clock_t start = clock();

        for (int i = 1; i <= 5; i++) {
            sprintf(buffer, "Data %d", i);
            write(fd[1], buffer, strlen(buffer) + 1);
            printf("Producer: %s\n", buffer);
        }

        clock_t end = clock();

        close(fd[1]);
        wait(NULL);

        double time_taken =
```

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
    perror("fork");
    return 1;
}

if (pid > 0) {
    // Parent - Producer
    close(fd[0]);

    clock_t start = clock();

    for (int i = 1; i <= 5; i++) {
        sprintf(buffer, "Data %d", i);
        write(fd[1], buffer, strlen(buffer) + 1);
        printf("Producer: %s\n", buffer);
    }

    clock_t end = clock();

    close(fd[1]);
    wait(NULL);

    double time_taken =
        (double)(end - start) / CLOCKS_PER_SEC;

    printf("Communication time: %f seconds\n", time_taken);
}
else {
    // Child - Consumer
    close(fd[1]);

    for (int i = 1; i <= 5; i++) {
        read(fd[0], buffer, sizeof(buffer));
        printf("Consumer: %s\n", buffer);
    }

    close(fd[0]);
}

return 0;
}
```

```
keerthi_priya@LAPTOP-OSGV × + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd practical5
keerthi_priya@LAPTOP-OSGVCE7A:~/practical5$ nano producer_consumer.c
keerthi_priya@LAPTOP-OSGVCE7A:~/practical5$ gcc producer_consumer.c -o producer_consumer
keerthi_priya@LAPTOP-OSGVCE7A:~/practical5$ ./producer_consumer
Producer: Data 1
Producer: Data 2
Producer: Data 3
Producer: Data 4
Producer: Data 5
Consumer: Data 1
Consumer: Data 1
Consumer: Data 1
Consumer: Data 1
Consumer: Data 1
Communication time: 0.000084 seconds
keerthi_priya@LAPTOP-OSGVCE7A:~/practical5$
```

TASK2: Develop a program that executes the equivalent of the shell command: `ls -l grep` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    int fd[2];
    pid_t p1, p2;

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    // First child: ls -l
    p1 = fork();

    if (p1 == -1) {
        perror("fork");
        return 1;
    }

    if (p1 == 0) {
        // Redirect stdout to pipe
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);

        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }
}
```

```
keerthi_priya@LAPTOP-OSGV × + ▾
GNU nano 8.7.1
    perror("execlp");
    exit(1);
}

// Second child: grep ".c"
p2 = fork();

if (p2 == -1) {
    perror("fork");
    return 1;
}

if (p2 == 0) {
    // Redirect stdin from pipe
    close(fd[1]);

    dup2(fd[0], STDIN_FILENO);

    close(fd[0]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp");
    exit(1);
}

// Parent
close(fd[0]);
close(fd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

|   return 0;
}
```

```
keerthi_priya@LAPTOP-OSGV × + ▾
keerthi_priya@LAPTOP-OSGVCE7A:~$ nano task2.c
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc task2.c -o task2
keerthi_priya@LAPTOP-OSGVCE7A:~$ ./task2
drwxr-xr-x  2 keerthi_priya keerthi_priya  4096 Aug  3 04:46 ProcessTask
drwxr-xr-x  2 keerthi_priya keerthi_priya  4096 Aug 22 06:00 practical3
drwxr-xr-x  2 keerthi_priya keerthi_priya  4096 Aug 17 05:34 practical4
drwxr-xr-x  2 keerthi_priya keerthi_priya  4096 Aug 26 10:08 practical5
-rw-r--r--  1 keerthi_priya keerthi_priya  1015 Aug 26 10:09 task2.c
keerthi_priya@LAPTOP-OSGVCE7A:~$ |
```

